APMA 3150: From Data to Knowledge
Final Research Project

Summer 2026 Session I

1 Description

This Final Research Project is a group project worth 20% of the course grade. In this project, first you will find a dataset from a public source. A few sources are the *UsingR* library, government websites, corporate websites, or any resource from the following:

- Google Dataset Search
- Kaggle
- UC Irvine Machine Learning Repository
- Open Data Portals
- GitHub: Example 1, Example 2
- Data at FiveThirtyEight
- Flowing Data (data visualization newsletter) datasets

Then examine the data using any of the tools learned in this class and find the "story" in the dataset. A good project will generate plots, try multiple techniques, and draw conclusions based upon these plots. If you use a dataset that has already been analyzed and discussed in some other source, then try to find a related dataset to run some new analyses that will yield a "new" story, or try some other statistical techniques on the dataset to find a new story.

Make sure to provide the website link showing where your dataset came from and indicate whether prior analysis has already been done on this dataset. If so, was there an R program that you started with? Or was there a program in some other language that you adopted? If so, what were your extensions? Simply repeating someone else's work in R is insufficient.

All projects should involve a publicly available dataset and include at least two (the more the merrier) of the following elements: simulation, confidence interval, hypothesis testing, linear regression, one-way/two-way ANOVA, logistic regression, tree model, clustering, PCA, etc.

2 Submission requirements

Please submit the following two items to the corresponding Canvas assignments:

1. Recorded presentation video named:
   Lastname1_Lastname2_Lastname3_...._Final_Project_Presentation.mp4.
2. Final written report named:
   Lastname1_Lastname2_Lastname3_...._Final_Project_Report.pdf.
   The report should be self-contained and not require the reader to execute your R program to understand the significance of your findings. There is no maximum page limit, but the report should be at least 4 pages, not including the references. The graphs and tables should be included, and references should be listed. Font size: 11 points; 1-inch margins all around; line spacing: 1.5. The report should include the following components:
   (a) Abstract (10%):
   Explain why your work is important, describe how you researched your problem, offer your conclusions, and highlight major points (less than 300 words).
   (b) Dataset (10%):
   Describe your dataset: Who made it? Why? How? When? (Remember that people create data!) How did you find it? What data does it contain? Where was it collected? How does this dataset's creation affect what you can learn from it?
   In other words, the reader should not be required to go to the reference to understand the dataset or the rest of the report. References must be included. If the dataset is part of a library, provide the name of the library and the name of the dataset in your reference.

(c) Research question/problem (10%):
What do you want to find out? Why is this an important topic that deserves to be studied?
What do scholars already know about this question? Was this dataset analyzed previously? If so, using what techniques? What were the previous findings? Provide references. References must be cited within the text. Discuss at least one reliable source, such as a paper describing a similar study to yours. Include a citation and a reference for every source.

(d) Research methods (15%):
How does this dataset relate to your research question?
How did you analyze your data? Why? Explain how your techniques were appropriate for your question and for your data. In a table, summarize which techniques produced meaningful findings and which did not. Which of these applied techniques were new relative to prior work, and which were repetitions of prior work?

(e) Project plan (15%):
How did you organize your work? How often did your team meet? Who led the tasks of organizing meetings, documenting your work, finding sources, writing your deliverables, etc.? Did you take turns doing these roles?
Include a timeline of your tasks and when you worked on them (including your presentation).
What skills did you need to learn to do this project well?

(f) Analyze data and results (30%):
Did you visualize the data well? From the visualizations, how did you interpret the possible results for the data?
Were the assumptions for applying the techniques checked? How did you interpret the results from applying the techniques? Did they reinforce your interpretation from the visualizations?
Graphs and tables should be embedded within the text and numbered. There should be text explaining each graph and table. Carefully select which graphs to include. Randomly including a whole lot of graphs with no explanations is not appropriate. Add x-axis and y-axis labels in your graphs. Add captions for the graphs and tables.

(g) Summary or conclusions (10%):
Briefly describe the story extracted from the dataset (less than 500 words).